higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

4 August 2023

This marking guideline consists of 6 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	The sustained departure of the control point from the desired value after a step of load change has taken place		
	1.1.2	The point of high velocity and low pressure		
	1.1.3	The taking in of a fluid to fill the gaps in a solid		
	1.1.4	The unit of dynamic or absolute viscosity		
	1.1.5	The mass of a substance relative to its volume		
			(5 × 2)	(10)
1.2	1.2.1	Saturation temperature		
	1.2.2	Faraday's law		
	1.2.3	Newtonian fluids		
	1.2.4	Flange taps		
	1.2.5	Feedback control		
			(5 × 1)	(5)
1.3	1.3.1	False		
	1.3.2	True		
	1.3.3	False		
	1.3.4	False		
	1.3.5	True		
			(5 × 1)	(5)
				[20]

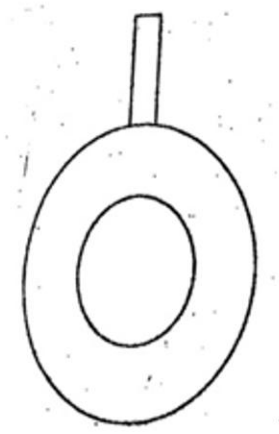
TOTAL SECTION A: 20

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SECTION B

QUESTION 2: FLOW MEASUREMENTS

2.1 2.1.1



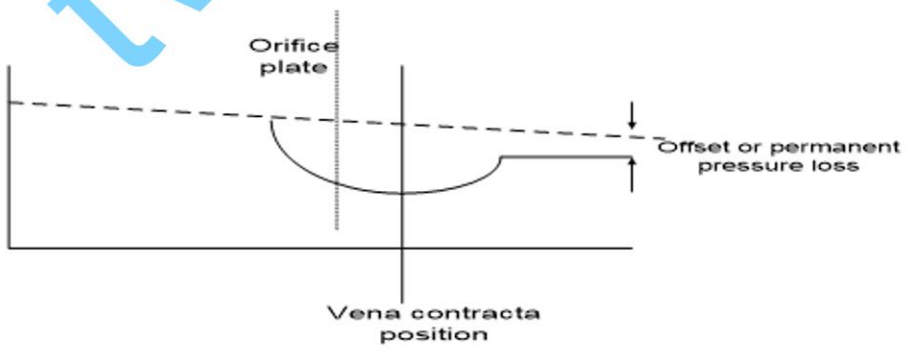
✓✓

2.1.2



✓✓

2.1.3



✓✓

(3 × 2) (6)

2.2 $Q = 8/15 \tan\frac{\theta}{2} \sqrt{2gH^5}$

$0,7 = 0,533 \times \tan 50/2 \sqrt{2gH^5}\surd$

$0,7 = 0,533 \times 0,466 \times \sqrt{2} \times 10 \times H^5\surd$

$2,8183 = \sqrt{2} \times 10 \times H^5\surd$

$1,679 = 20H^5 \surd$

$0,08395 = H^5\surd$

$H = 0,6093\ m\surd$ (5)

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2.3 Advantages:

- Measure liquids, vapours and gases
- Connection does not freeze
- Can be used for high temperatures
- Less sensitive to change in Reynolds number
- Very accurate (Any 3 × 1) (3)

Disadvantages:

- Can only be zeroed under no-flow conditions
- In-line mounting required (2)

2.4 The output for a target flow meter is either a electric or a pneumatic signal,✓ whereas the drag body flowmeter has only an electrical output signal.✓ (2)

2.5

- Process fluid should be defined by its properties
- The viscosity of the fluid over the operating temperature ranges
- The linear range
- The temperature and the pressure over which the meter will be operated
- The process connections
- The signal transmission system
- The type of totaliser and rate of flow indicator (Any 4 × 1) (4)

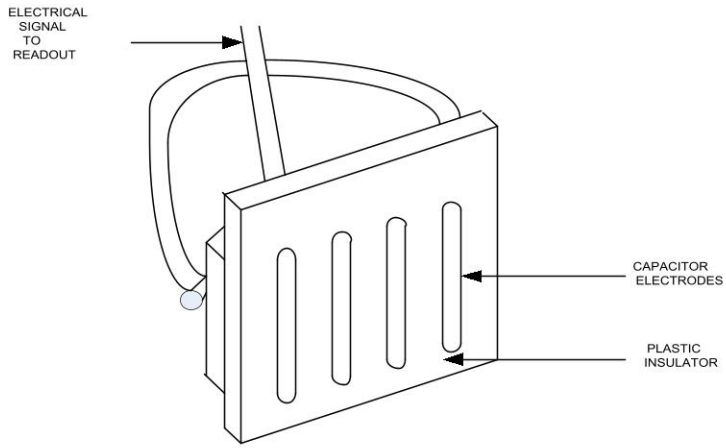
[22]

QUESTION 3: DENSITY, HUMIDITY AND VISCOSITY

3.1

- Hydrostatic methods
- Buoyancy methods
- Radiation methods (3)

3.2



- This instrument consists of stainless-steel electrodes mounted on a plastic isolator.✓
- These electrodes form the plates of the capacitor✓ and the dielectric will be influenced by the amount of vapour present in the substance being measured.✓
- The change in capacitance can be measured by a capacitor Wheatstone bridge or an LC resonant circuit.✓

(3 marks for the sketch and 4 marks for the operation) (7)

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3.3 3.3.1 Sliding-plate viscometer (2)

3.3.2
- A = Servo motor
- B = Potentiometer signal to recorder
- C = Insulating coupling
- D = Micrometer
- E = Flag
- F = Sliding plate
- G = Constant-temperature bath
- H = Sample
- I = Stationary plate
- J = Weights
- K = Beam (11)

[23]

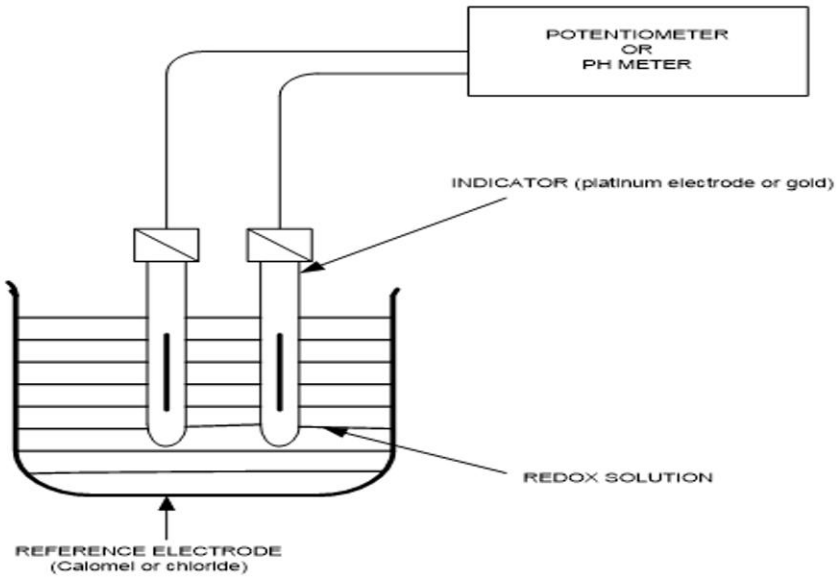
QUESTION 4: pH AND CONDUCTIVITY MEASUREMENT

4.1 $pH = -\log_{10}[H^+]$
$6,85 = -\log_{10}[H^+]$✓
$H^+ = 10^{-6,85}$ ✓✓
$= 1,413 \times 10^{-7}$✓
There are $1,413 \times 10^{-7}$ mole/hydrogen ions present in the solution. (4)

4.2 Conductivity differs from pH because all ions present contribute to the conductivity,✓ whereas pH is affected only by hydrogen ions.✓ (2)

4.3
- By utilising manual compensators that consist of a variable and a fixed resistor in series
- By using automatic compensators that consist of a thermistor temperature sensor (2 × 2) (4)

4.4



(1 mark for the sketch and 4 marks for each label) (5)

[15]

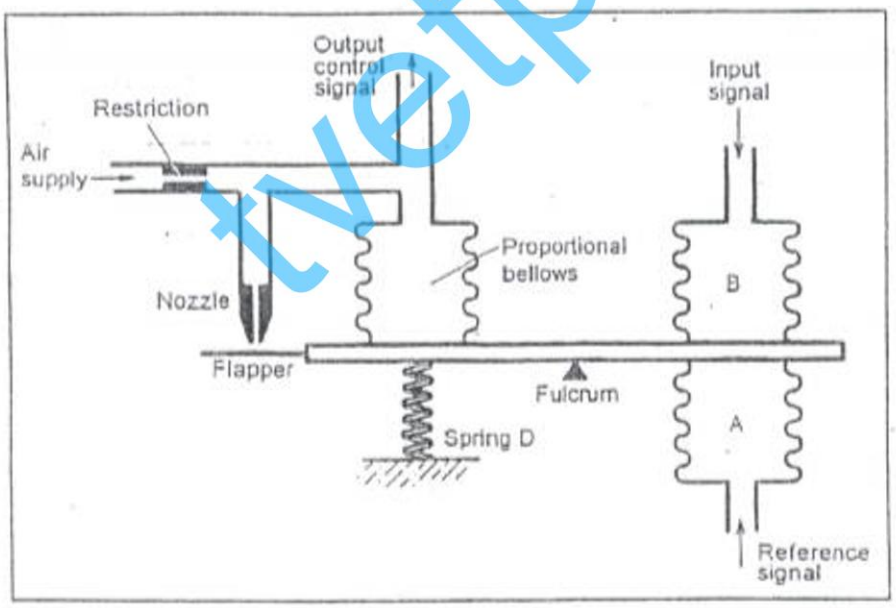
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QUESTION 5: AUTOMATIC CONTROL

5.1 5.1.1 On-off pneumatic controller (2)

5.2.2
- The controller has no feedback and operates indirectly to the deviation.✓
- When the error increases, the flapper is moved towards the nozzle.✓
- The pressure will build up and cause the relay to de-amplify the error to the control valve,✓ which in turn will shut the air supply down to rectify the deviation.✓
- The error will decrease pass the set point and move the flapper away from the nozzle, again resulting in a drop in pressure.✓
- The relay will amplify the error to the control valve, which will open up the control valve to rectify the deviation.✓
- This action will be repeated and oscillation above and below the set point will occur.✓ (7)

5.2



(2 marks for the correct sketch an half a mark for each label) (7)

5.3
- Operational amplifier
- Wheatstone bridge
- Linear variable differential transformer
- Linear variable capacitive transducer (4)

[20]

TOTAL SECTION B: 80
GRAND TOTAL: 100

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